Square Star Pattern :-

column = n (Inner loop)

* for n = 5 :- rows = n (outer loop)

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

* logic :-

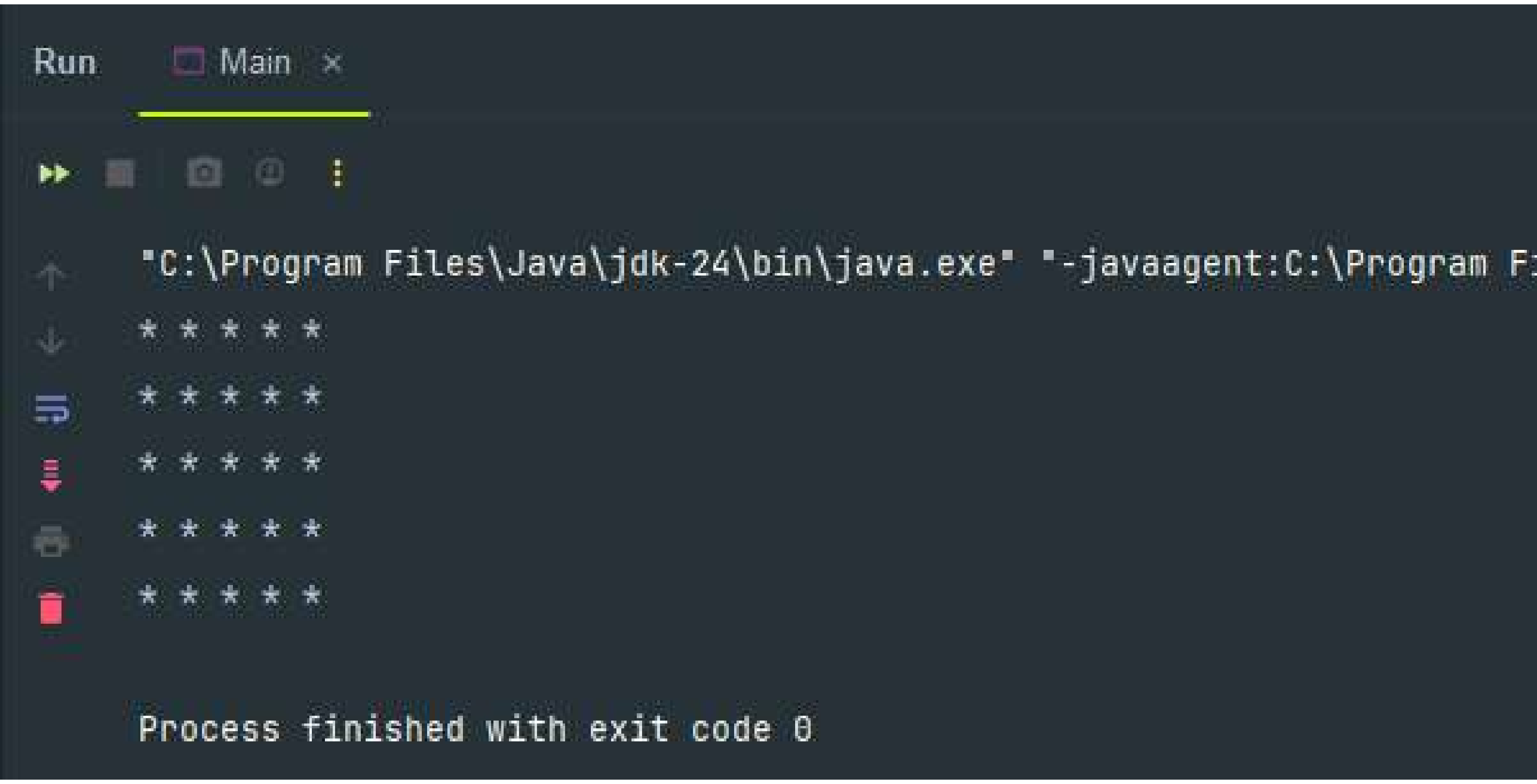
i (Row)		star
1	$\longrightarrow$	5
2	$\longrightarrow$	5
3	$\longrightarrow$	5
4	$\longrightarrow$	5
5	$\longrightarrow$	5

* Pseudo code :-

```
Repeat n times
    Repeat n times
        print *
    Move to the next line
```

* Code :-

```
int n=5
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        System.out.print("* ");      ⟶ print *
    }
    System.out.println();            ⟶ move to next line.
}
```


Run
Main
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Fi
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Process finished with exit code 0

⇒ Hollow Square Star Pattern :-

```
              j=1        j=n
              ↓          ↓
* for n=5,  i=1 ──→ * * * * *
                    *       *
                    *       *
                    *       *
            i=n ──→ * * * * *
```

*** logic :-**

Row (i)	Star
• first row ⟶	all stars
• Last row ⟶	all stars
• first column ⟶	all stars
• Last column ⟶	all stars
• Remaining positions ⟶	spaces

*** Pseudo code :-**

```
for i=1 to n
    for j=1 to n
        if i==1 OR i==n OR j==1 OR j==n
            print "*"
        else
            print " "
    print new line
```

*** Code :-**

```
int n=5;
for (int i=1; i<=n; i++){
    for (int j=1; j<=n; j++){
        if (i==1 || i==n || j==1 || j==n){
            System.out.print("*");
        }else{
            System.out.print(" ");
        }
    }
    System.out.println();
}
```

Run Main

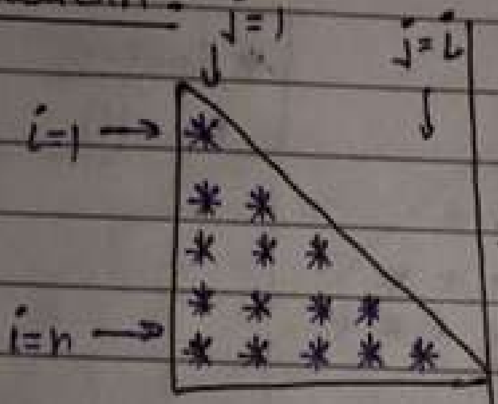
```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
* * * * *
*       *
*       *
*       *
* * * * *

Process finished with exit code 0
```

Date

2013

3) Triangle Pattern :-

* for n=5,



* Logic :-

Row(i)	Star
1	1
2	2
3	3
4	4
5	5

* Pseudo code :-

```
for i=1 to n
    for j=1 to i
        print "*"
    print new line
```

* Code :-

```
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}
```


Run
Main
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Com
*
* *
* * *
* * * *
* * * * *
Process finished with exit code 0

4) Inverted Triangle Pattern:-

* for n=5:

```
        j=1
i=n →  * * * * *
       * * * *
       * * *
       * *
i=1 →  *
```

* Logic:-

Row (i)		Star
5	→	5
4	→	4
3	→	3
2	→	2
1	→	1

* Pseudo code:-

```
for i=1 to n
    for j=1 to (n-i+1)
        print "*"
    print new line
```

* Code:-

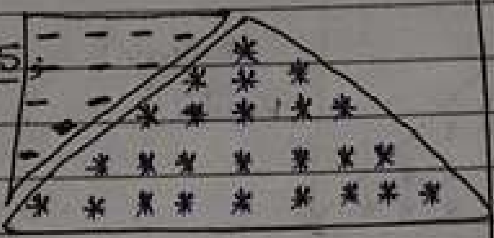
```java
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n-i+1; j++){
        System.out.print("* ");
    }
    System.out.println();
}
```


Run
Main
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
* * * * *
* * * *
* * *
* *
*
Process finished with exit code 0

Date 2013

5) Pyramid Pattern:-

* for n=5:



* logic:-

n-i (2*i-1)

Row(i)	spaces	Star
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9

* Pseudo code:-

```
for i=1 to n
    for space=1 to n-i
        print space
    for star=1 to 2*i-1
        print "*"
    print new line
```

* Code:-

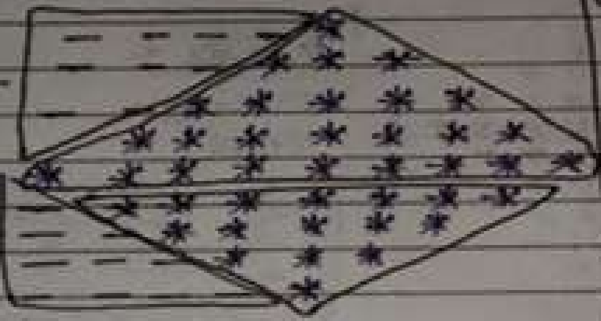
```
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n-i; j++){
        System.out.print
        (" ");
    }
    for(int k=1; k<=(2*i-1);
    k++){
        System.out.print("*");
    }
    System.out.println();
}
```

Run Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community
    *
   ***
  *****
 *******
*********

Process finished with exit code 0
```

6) Diamond Pattern:-

*- for n=5,



*** Logic :-**

① Pyramid

Row(i)	Spaces	Star
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9

② Inverted Pyramid
↓
reverse version of Pyramid

*** Code :-**

```
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n-i; j++){
        System.out.print(" ");
    }
    for(int k=1; k<=2*i-1; k++){
        System.out.print("*");
    }
    System.out.println();
}

for(int i=n-1; i>=1; i--){
    for(int j=1; j<=n-i; j++){
        System.out.print(" ");
    }
    for(int k=1; k<=2*i-1; k++){
        System.out.print("*");
    }
    System.out.println();
}
```

*** Pseudo code :-**

① Upper half,
- for i=1 to n
  - print spaces(n-i)
  - print star(2*i-1)
  - new line

② Lower half,
- for i=n-1 down to 1
  - print spaces (n-i)
  - print stars(2*i-1)
  - new line

Run   Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *

Process finished with exit code 0
```

⑦ Half diamond Pattern :-

*** for n=5 :**

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

*** Logic :-**

① Triangle

Row(i)	Stars
1	1
2	2
3	3
4	4
5	5

② Inverted triangle
↳ It reverse of triangle

*** code :-**

```
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}

for(int i=n-1; i>=1; i--){
    for (int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}
```

*** Pseudo code :-**

1) upper half;

for i=1 to n
  print i stars
  new line

2) lower half;

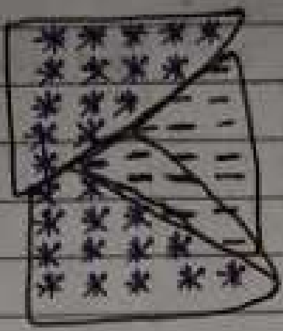
for i=n-1 down to 1
  print i stars
  new line

Run   Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\lib\idea_rt.
*
**
***
****
*****
****
***
**
*

Process finished with exit code 0
```

8> Half inverted diamond pattern:-

* for n=5:



* Logic:-
① Inverted triangle
+
② triangle

* Code:-

```
int n=5;
for(int i=n; i>=1; i--){
  for(int j=1; j<=i; j++){
     System.out.print("*");
  }
  System.out.println();
}

for(int i=2; i<=n; i++){
  for(int j=1; j<=i; j++){
     System.out.print("*");
  }
  System.out.println();
}
```

* Pseudo code:-
① Upper half
for i=n down to 1
print i stars
new line

② Lower half.

for i=2 to n
print i stars
new line

Run   Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\lib\idea_rt.
*****
****
***
**
*
**
***
****
*****

Process finished with exit code 0
```

Date

1) Square Number Pattern:

*- for n=5:

```
         ↓j=1 - - ↓j=n
i=1 →    1 1 1 1 1
 |       1 1 1 1 1
 |       1 1 1 1 1
 |       1 1 1 1 1
i=n →    1 1 1 1 1
```

*Logic:

Row(i)		print 1
1	→	5
2	→	5
3	→	5
4	→	5
5	→	5

*Code:-

```java
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        System.out.print("1 ");
    }
    System.out.println();
}
```

*Pseudo code:

```
for i=1 to n
    for j=1 to n
        print 1
print new line
```


Run
Main
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
Process finished with exit code 0

print new line

(2) Incrementing Square Pattern:-

* for n=5

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

* Logic :-

Row(i)	print
1 →	1 to 5
2 →	"
3 →	"
4 →	"
5 →	"

* Code:-

```
int n=5;
for(int i=1; i<=n; i++){
  for(int j=1; j<=n; j++){
    System.out.print(j+" ");
  }
  System.out.println();
}
```

* Pseudo code:-

```
for i=1 to n
   for j=1 to n
       print j
   print new line
```

Run   Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Communit
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

Process finished with exit code 0
```

3) Incrementing Right Triangle Pattern:-

*** for n=5,**

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

*** Logic:-**

Row(i)		print
1	→	1
2	→	1 to 2
3	→	1 to 3
4	→	1 to 4
5	→	1 to 5

*** Code:**

```
        int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++) {

        System.out.print(j+" ");
    }
    System.out.println();
}
```

*** Pseudo code:**

```
For i=1 to n
  for j=1 to i

     print j

  print new line
```

Run Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Process finished with exit code 0
```

4) Inverted diamond Pattern:

*** for n=5,**

```
33333
 4444
  555
   66
    7
```

*** Logic:-**

- Spaces = i ⇒ increase
- Number = 3+i ⇒ increase
- Count = n-i ⇒ decrease

*** Code:**

```
int n=5;
int num=3

for(int i=0; i<n; i++){
  for(int j=0; j<i; j++){
     System.out.print(" ");
  }

  for(int k=0; k<n-i; k++){
     System.out.print(num+" ");
  }
  num++;
  System.out.println();
}
```

*** Pseudo code:-**

```
num=3
For i=0 to n-1
   print i spaces
   print num (n-i) times
  num++
  print new line
```

Run Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\l
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7

Process finished with exit code 0
```

Date 2013

9) Sandwich Pattern:

*

```
3 3 3 3 3
 4 4 4
   5
 4 4 4
3 3 3 3 3
```

*** Logic :-**

- Number : 3 → 4 → 5 → 4 → 3
- Count : 5 → 3 → 1 → 3 → 5

*** Code :**

```
for(int i=0; i<3; i++){
  for(int j=0; j<i; j++){
     System.out.print(" ");
  }
  for(int k=0; k<5-(2*i); k++){
     System.out.print((3+i)+" ");
  }
  System.out.println();
}

for(int i=1; i>=0; i--){
  for(int j=0; j<i; j++){
     System.out.print(" ");
  }
  for(int k=0; k<5-(2*i); k++){
     System.out.print((3+i)+" ");
  }
  System.out.println();
}
```

*** Pseudocode :**

```
For i=0 to 2
   print i spaces
   print (3+i) (5-2*i) times
   print new line

For i=1 downto 0
   print i spaces
   print (3+i) (5-2*i) times
   print new line
```

Run Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\lib\i
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

Process finished with exit code 0
```

6) Incrementing Diamond Pattern:-

*

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

* Logic :

Number: 3 → 4 → 5 → 6 → 7
↓
6 → 5 → 4 → 3

Count = No. of elements in row

* Code :-

```java
for(int i=1; i<=5; i++){
    int num=i+2;
  for(int j=1; j<=i; j++){
    System.out.print(num+" ");
  }
  System.out.println();
}

for(int i=4; i>=1; i--){
    int num = i+2;
  for(int j=1; j<=i; j++){
    System.out.print(num+" ");
  }
  System.out.println();
}
```

* Pseudo code:

```
For i=1 to 5
    print (i+2) i times
    print new lines

for i=4 down to 1
    print (i+2) i times
    print new line
```

Run   Main ×

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\l
    1
   2 2
  3 3 3
 4 4 4 4
5 5 5 5 5
 4 4 4 4
  3 3 3
   2 2
    1

Process finished with exit code 0
```